

A Novel Method for Text-to-Sign Language Generation Using Hierarchical Semantic Motion Units

1 Field of the Invention

The present invention relates to the field of human-computer interaction, assistive technologies, and artificial intelligence. More specifically, the invention concerns a method and system for converting natural language text into sign language representations using adaptive and hierarchical motion modeling.

2 Background and Prior Art

Existing text-to-sign language systems generally follow a pipeline consisting of text pre-processing, gloss generation, pose estimation, and sign animation. While effective, such systems rely on direct gloss-to-pose mappings, resulting in limited adaptability, poor generalization to unseen signs, and lack of personalization.

Furthermore, prior art systems generate deterministic sign motions without accounting for linguistic uncertainty, user context, or semantic decomposition of sign gestures. This limits interpretability, scalability, and usability in real-world scenarios, particularly for low-resource sign languages.

3 Problem Statement

There exists a need for a text-to-sign language generation system that:

- Generalizes to unseen or rare sign expressions
- Produces smooth, interpretable, and adaptive sign motions
- Separates linguistic intent from low-level motion execution
- Allows personalization and uncertainty-aware signing

4 Overview of the Proposed Method

The proposed invention introduces a hierarchical intermediate representation called **Semantic Motion Units (SMUs)**, which decouple linguistic semantics from physical motion generation.

Text → Gloss → Semantic Motion Units → Pose → Sign Output

This decomposition enables adaptive, scalable, and patentable sign language synthesis.

5 Detailed Pipeline Description

5.1 Stage 1: Text Input

The input is a spoken language sentence such as:

“I am going to school tomorrow.”

This sentence follows spoken-language grammar and includes auxiliary words not present in sign language.

5.2 Stage 2: Text Preprocessing

Natural Language Processing techniques are applied to clean and structure the input text.

Operations include:

- Tokenization
- Removal of auxiliary verbs
- Identification of subject, action, and temporal context

This step prepares the input for linguistic transformation without altering semantic meaning.

5.3 Stage 3: Text-to-Gloss Translation

The preprocessed text is converted into a gloss sequence that follows sign language grammar.

Text: I am going to school tomorrow

Gloss: ME SCHOOL GO TOMORROW

This is treated as a sequence-to-sequence translation problem and implemented using neural language models.

5.4 Stage 4: Semantic Motion Unit Generation (Core Novelty)

Instead of directly converting gloss tokens into pose sequences, each gloss is decomposed into one or more **Semantic Motion Units (SMUs)**.

Each SMU represents a linguistically meaningful motion primitive, defined by:

- Handshape category
- Hand orientation
- Motion trajectory
- Spatial location
- Non-manual markers (optional)

SMUs form an intermediate symbolic–continuous representation bridging language and motion.

5.5 Stage 5: Temporal Motion Planning

The SMU sequence is passed through a temporal planner that enforces:

- Inter-unit smoothness
- Velocity continuity
- Coarticulation constraints

This step ensures that the final sign motion is natural and human-like.

5.6 Stage 6: Pose Generation

The temporally refined SMUs are converted into pose sequences consisting of body, hand, and facial keypoints.

The output format is:

$$T \times J \times (x, y, z)$$

where T denotes time steps and J denotes joint indices.

5.7 Stage 7: Sign Rendering

The generated pose sequence is rendered as:

- A 3D avatar animation, or
- A skeletal pose visualization

This representation can be used in assistive interfaces, educational tools, or communication systems.

6 Adaptive and Confidence-Aware Extensions

The system optionally integrates confidence scores associated with SMUs. Low-confidence motion units trigger:

- Slower execution
- Increased motion amplitude
- Emphasized articulation

This improves intelligibility and user trust.

7 Data Resources and Training

The invention may be implemented using publicly available datasets, including:

- PHOENIX-2014-T (text-gloss-pose)
- How2Sign (multimodal sign language dataset)
- WLASL (isolated sign dataset)

Pose data may also be extracted using pose estimation frameworks for training purposes.

8 Advantages of the Proposed Method

The proposed system provides:

- Improved generalization to unseen signs
- Modular and interpretable sign generation
- Support for personalization and adaptation
- Reduced dependency on large annotated datasets

9 Patent Novelty Summary

The novelty of the invention lies in:

- The introduction of Semantic Motion Units as an intermediate representation
- Hierarchical separation of linguistic and motion components
- Confidence-aware and adaptive motion synthesis
- A structured method for scalable sign language generation

10 Conclusion

This invention presents a structured, extensible, and patentable approach to text-to-sign language generation by introducing a novel intermediate motion representation that bridges linguistic semantics and physical articulation. The method advances the state of the art in assistive communication systems and sign language synthesis.